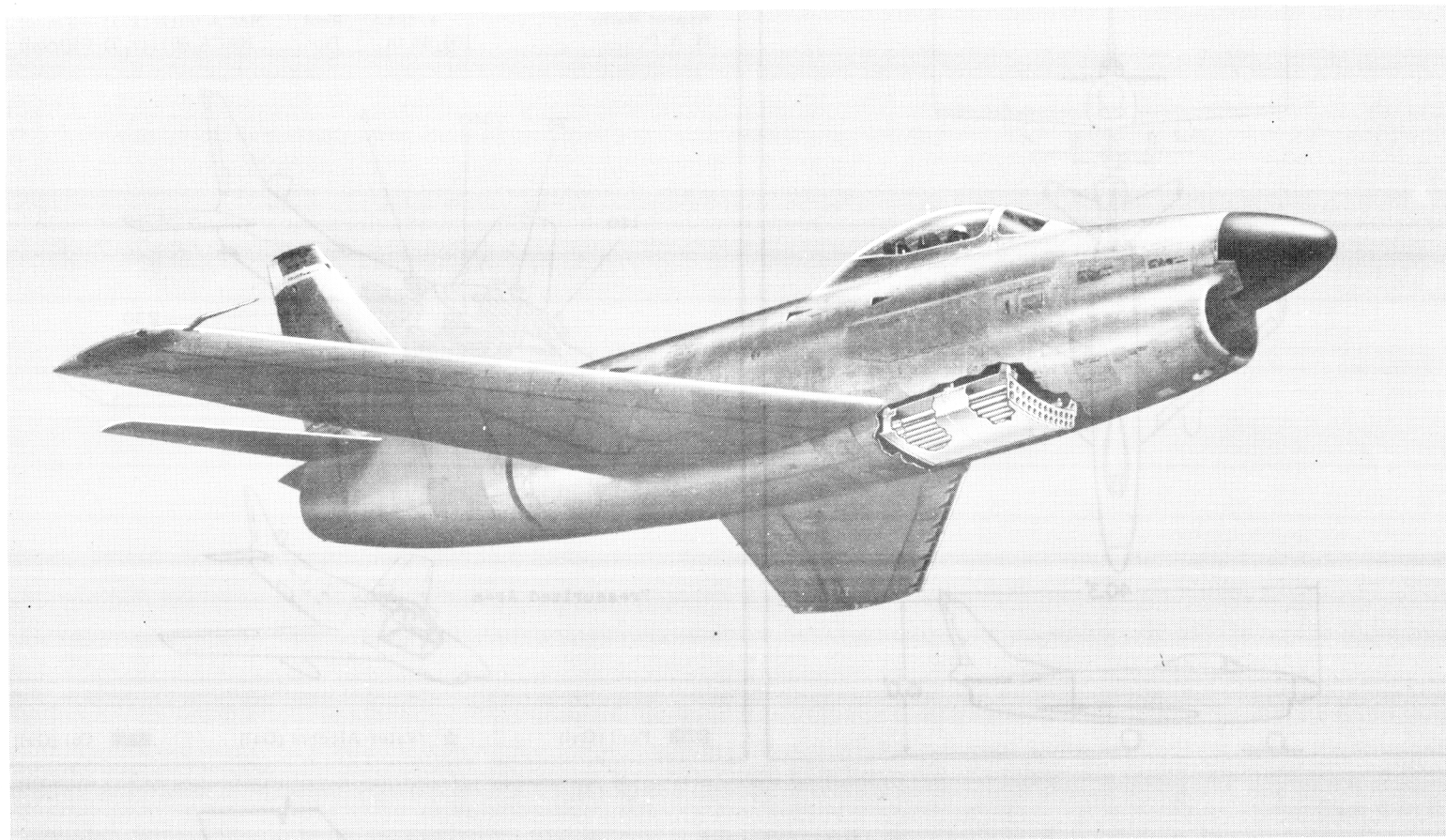


U N C L A S S I F I E D

*H1
F-86L/char*

SERVICE



Standard Aircraft Characteristics

BY AUTHORITY OF
THE SECRETARY
OF THE AIR FORCE

F-86 L
SABRE
North American

ONE J47-GE-33
GENERAL ELECTRIC

22 SEP 58

U N C L A S S I F I E D

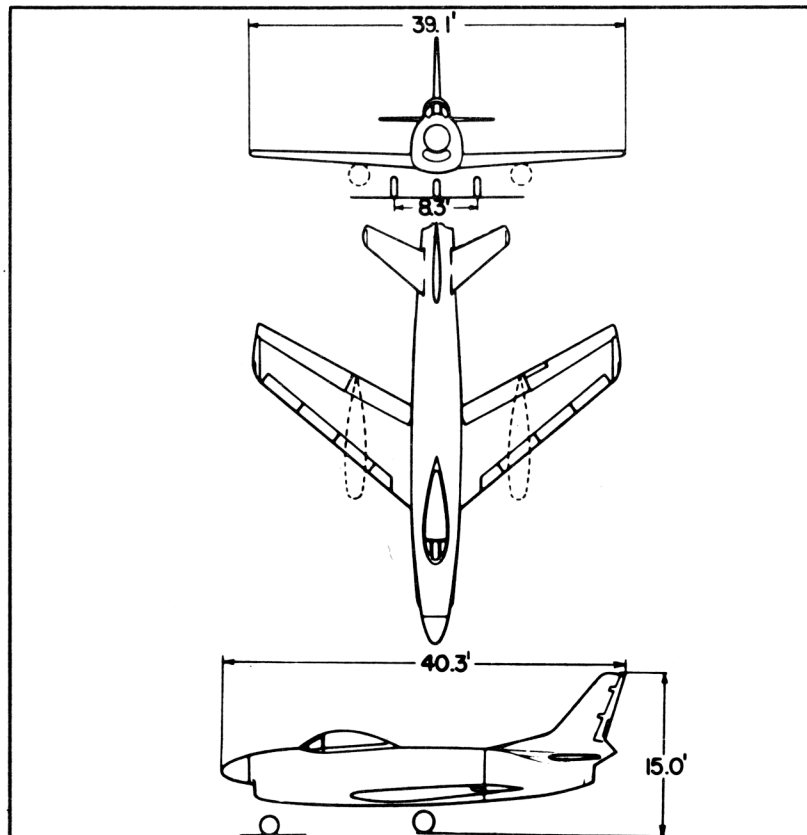
F-86L

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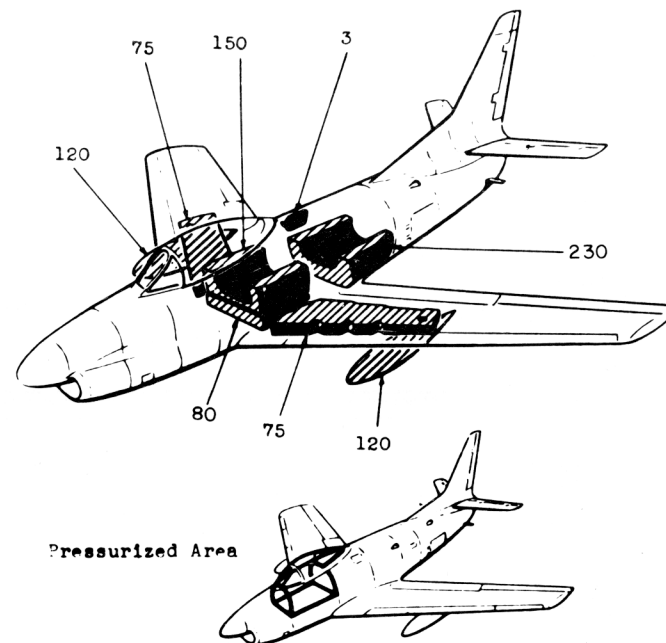
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Wing Area 313.37 sq ft Wing Section
 Aspect Ratio 4.883 Root . . NACA 0012(11.1)-64(mod)
 M.A.C. 100.66 in Tip . . . NACA 0011(9.7)-64(mod)

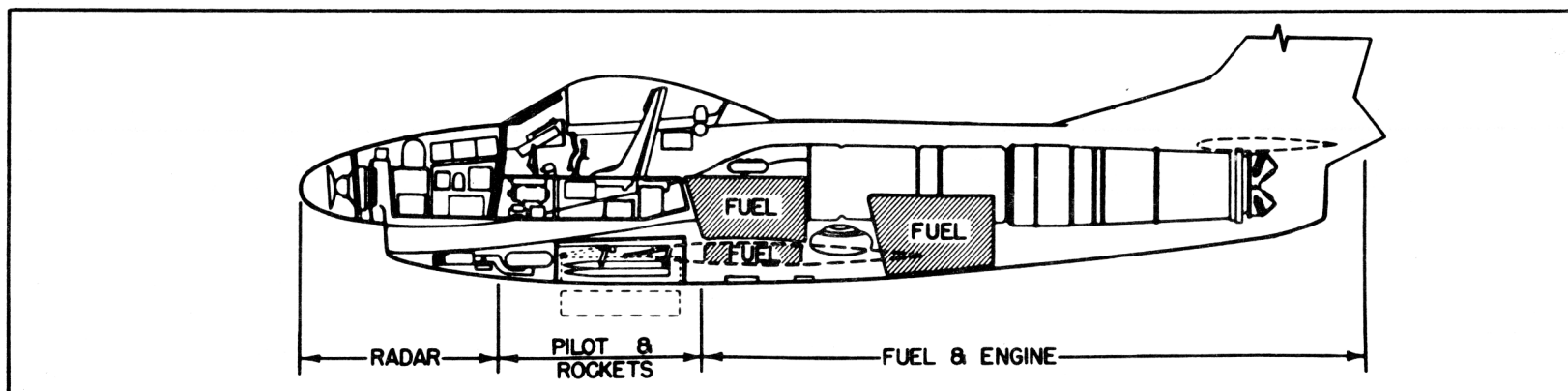


Pressurized Area

Fuel (Gal)

☆ Water Alcohol (Gal)

Oil (Gal)



F-86L

U N C L A S S I F I E D

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POWER PLANT

Nr & Model . . . (1) J47-GE-33
 Mfr . . . General Electric
 Engine Spec Nr . . . E-644
 Type . . . Axial
 Length . . . 226.5"
 Diameter . . . 37.0"
 Weight(dry) . . . 3275 lb
 Tail Pipe . . . Auto, Variable Area
 Augmentation . . . Afterburning

ENGINE RATINGS

S. L. Static LB - TEMP - MIN

Max: *7650 - 685°C - 15

Mil: 5550 - 685°C - 30

Nor: 5100 - 635°C - Cont

* With afterburner operating

7950 RPM remains constant for normal power and higher.

DIMENSIONS

Wing
 Span . . . 39.1'
 Incidence (root) . . . 1°
 (tip) . . . -1°13'
 Dihedral . . . 3°
 Sweepback (25% chord) . . . 35°51'
 Length . . . 40.3'
 Height . . . 15.0'
 Tread . . . 8.3'

Mission and Description

Navy Equivalent: None

Mfr's Model: NA-190, -201

The principal mission of the F-86L is the interception and destruction of hostile aircraft in flight during day or night and in all types of weather.

The airplane surface control system provide for: hydraulic power-operated irreversible controls, with artificial feel for the all-movable horizontal tail and ailerons, and a conventional pilot operated rudder.

The cockpit is provided with pressurization, heating and cooling, ejection seat, anti-G suit provisions, jettisonable hinged-type canopy and an F-5 automatic pilot.

High lift devices consist of electrically operated single slotted wing flaps and aerodynamically actuated wing leading edge slats. Speed brakes are also provided.

An E-4 all-weather rocket fire control system is incorporated to provide position information that enables the pilot to locate airborne targets, maneuver the airplane to launch an attack and accurately fire rockets.

Development

The F-86L aircraft are modified F-86D's. The modification consists of modernization of the electronic equipment and incorporation of the 6-3 slatted leading edge with 12 in. wing tip extensions for improved performance.

WEIGHTS

Loading	Lb	L. F.
Empty	13,822(A)	
Basic	13,847(A)	
Design	13,395	7.33
Combat(Point) *16,252		
Combat(Area) *17,160		
Max T.O.	↑ 20,276	2.00
Max in Flt	↑ 20,276	
Max Land	↑ 20,276	2.00

(A) Actual

* For Basic Mission

↑ Limited by mission

FUEL

Location	Nr Tanks	Gal
Wing	2	150
Fuselage	3	460
Wing, drop	2	240
	Total	850
Grade		JP-4
Specification		MIL-F-5624

OIL

Fuselage	1 (tot) 3
Grade	1005
Specification	MIL-L-6081

BOMBS

NONE

GUNS

NONE

ROCKETS

Nr	Size	Type	Location
24	2.75"	FFAR	Fuselage

Rockets are carried in a retractable launcher.

ELECTRONICS

UHF Command	AN/ARC-34
SIF	AN/APX-25
Glide Path	AN/ARN-31
Directional Range	AN/ARN-21
Marker Beacon	AN/ARN-12
Radar Ranging	AN/APG-37
Rocket Fire Computer	AN/APA-84
IFF	AN/APX-6A
Omni-Direct	AN/ARN-14
Intercom	AN/AIC-10
Data Link	AN/ARR-39

*Provisions only

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Loading and Performance—Typical Mission

C O N D I T I O N S			BASIC MISSION (INTERCEPT)		DESIGN MISSION (INTERCEPT)		FERRY RANGE
			POINT	AREA	POINT	AREA	
TAKE-OFF WEIGHT	(lb)	I	18,484	II 20,276	III 18,484	IV 20,276	V 19,884
Fuel at 6.5 lb/gal (grade JP-4)	(lb)		3965	5525	3952	5525	5525
Payload (rockets)	(lb)		432	432	432	432	None
Wing loading	(lb/sq ft)		59.0	64.8	59.0	64.8	63.4
Stall speed (power off)	(kn)		113	119	113	119	118
Take-off ground run at SL	(ft)		2450	3000	2450	3000	2870
Take-off to clear 50 ft	(ft)		3650	4400	3650	4400	4210
Rate of climb at SL	① (fpm)		11,100 ⑥	3400 ② ⑥	10,650	3300 ②	3420 ②
Time: SL to 40,000 ft	① ④ ⑤ (min)		7.0	—	7.0	—	—
Time: SL to service ceiling	① ⑤ (min)		19.1 ④	39.4 ②	9.2 4 / 40,000	36.5 ②	38.7 ②
Service ceiling (100 fpm)	① (ft)		49,600 ⑤	38,750 ② ⑤	47,950	34,000 ②	34,500 ②
COMBAT RANGE	③ (n. mi)		—	—	—	—	652
COMBAT RADIUS	③ (n. mi)		—	227	—	244	—
Average cruise speed	(kn)		—	478	—	458	477
Initial cruising altitude	(ft)		—	35,200	—	35,400	35,650
Final cruising altitude	(ft)		—	38,200	—	38,700	38,800
Total mission time	(hr)		—	1.11	—	1.33	1.42
TOTAL MISSION TIME	③ (hr)		0.56	—	0.69	—	—
Interception altitude	(ft)		48,250	—	45,000	—	—
COMBAT WEIGHT	(lb)		16,252	17,160	16,254	17,272	15,252
Combat altitude	(ft)		48,250	47,600	45,000	36,100	38,800
Combat speed	① (kn)		500	500	523	536	537
Combat climb	① (fpm)		500	500	1600	4500	4400
Combat ceiling (500 fpm)	① (ft)		48,250	47,600	48,250	47,550	49,400
Service ceiling (100 fpm)	① (ft)		49,400	39,400 ②	49,400	39,250 ②	41,350 ②
Max rate of climb at SL	① (fpm)		12,200	11,600	12,200	11,500	12,950
Max speed at S. L.	① (kn)		602	602	602	602	602
Basic speed at 40,000 ft	① (kn/ft)		535	534	—	—	540/35,000
LANDING WEIGHT	(lb)		15,417	15,700	14,927	15,083	15,258
Ground roll at SL	(ft)		2590	2640	2500	2540	2570
Ground roll (auxiliary brake)	⑦ (ft)		1600	1630	1580	1590	1600
Total from 50 ft	(ft)		3850	3900	3750	3780	3810
Total from 50 ft (auxiliary brake)	⑦ (ft)		2890	2950	2820	2840	2870

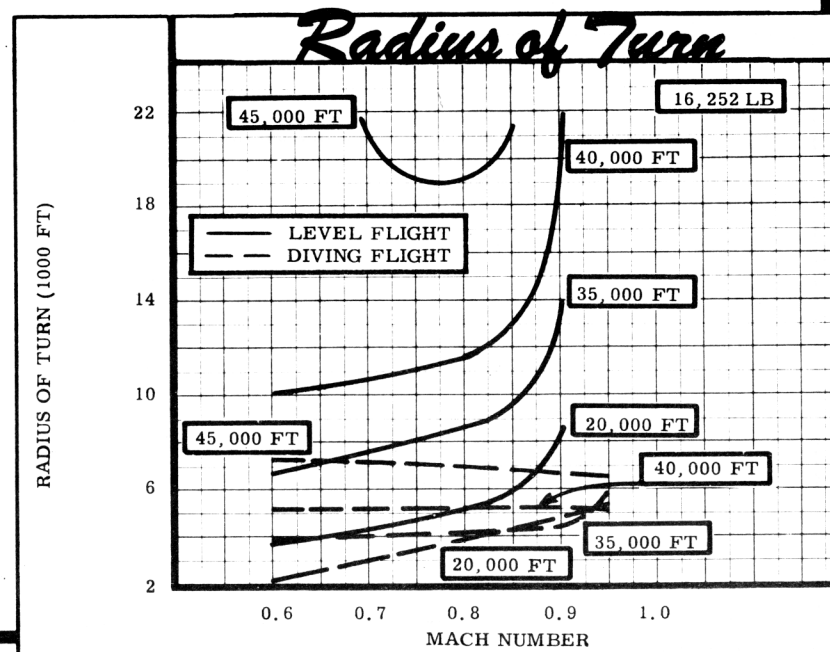
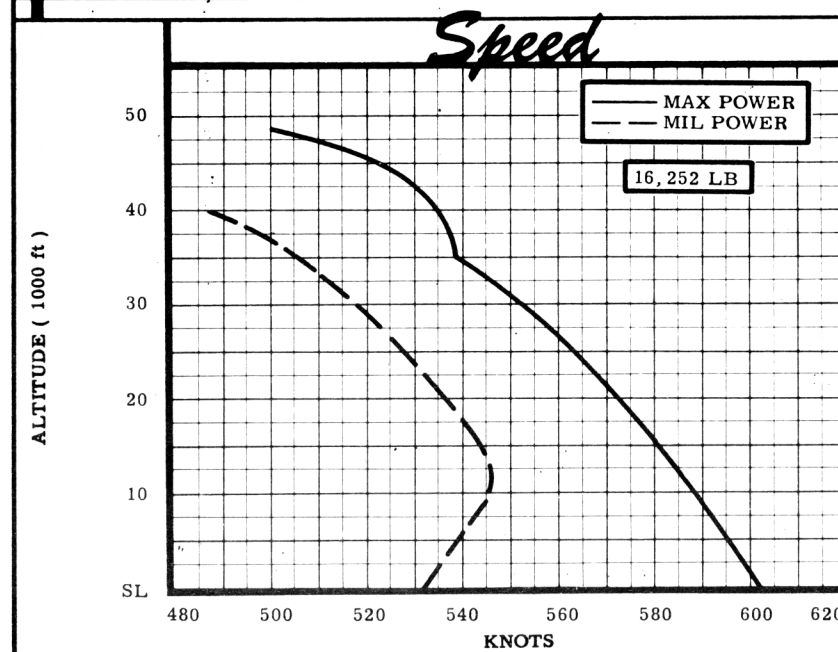
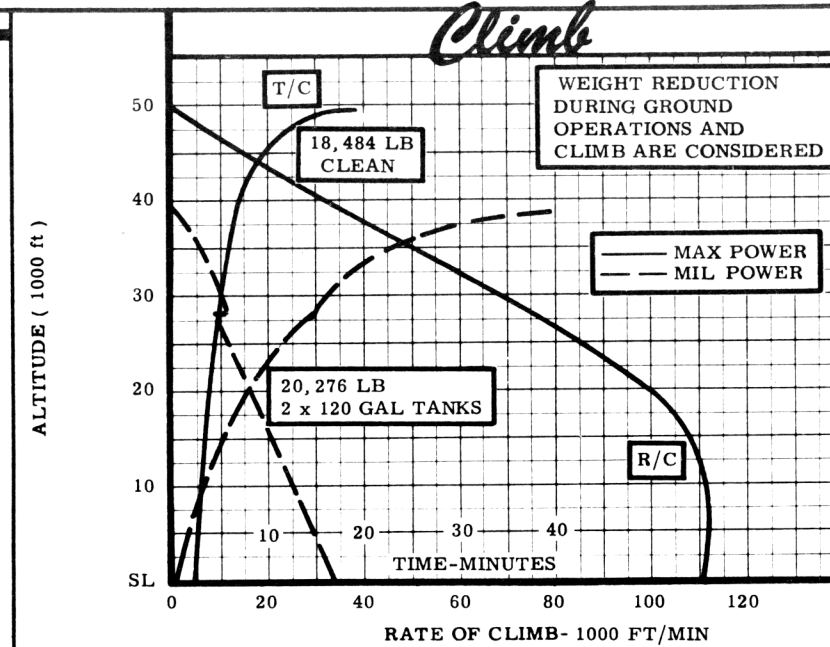
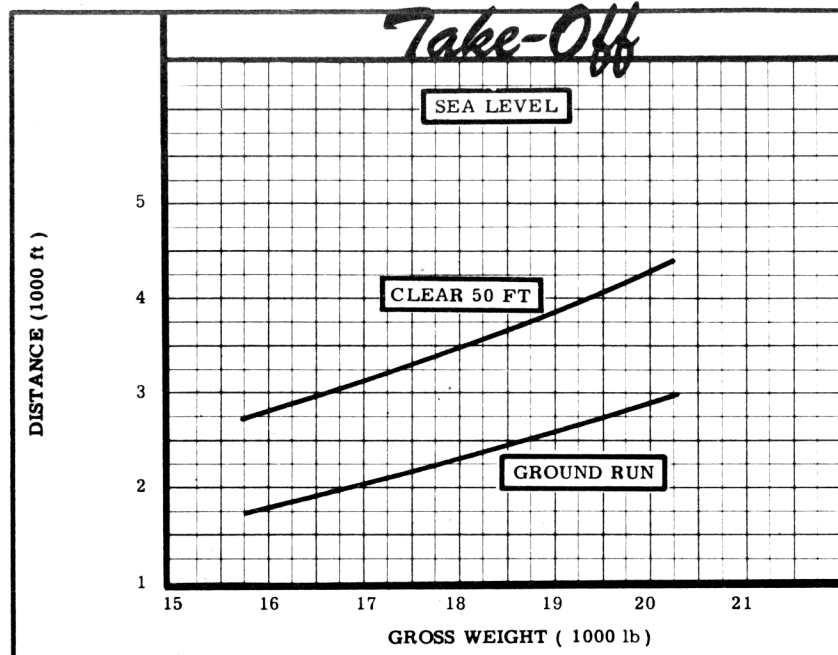
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- ① Maximum power
 ② Military power
 ③ Detailed descriptions of Radius and Range missions given on page 6.

- ④ Includes 2.0 minutes for take-off and acceleration to best climb speed
 ⑤ Allows for weight reduction during ground operation and climb

- ⑥ Allows for weight reduction during ground operation and acceleration to climb speed.
 ⑦ With 15.6 ft ring-slot drag chute.

PERFORMANCE BASIS:
 (a) Data source: Calculations based on flight test of F-86D
 (b) Performance is based on powers shown on page 6.



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N O T E S**FORMULA: POINT-INTERCEPT MISSION I**

Take-off and accelerate to best climb speed at maximum power, climb to combat ceiling with maximum power, combat for 5 minutes with fuel consumption based on maximum power at combat ceiling. Descend to 35,000 feet and loiter at speeds for maximum endurance until only reserve fuel remains. Fuel allowances include 2 minutes at normal power for starting engine and taxi; 1 minute at maximum power for take-off plus a reserve of 20 minutes loiter at sea level at speeds for maximum endurance. Total mission time includes all operational time from start of take-off run to end of loiter.

FORMULA: AREA-INTERCEPT MISSION II

Take-off and accelerate to best climb speed at maximum power, climb to cruise altitude with military power, cruise at cruise altitude at long range speeds, climb to combat ceiling with maximum power, combat 5 minutes at combat ceiling with maximum power, cruise back at cruise altitude at long range speeds. Range-free allowances include 2 minutes of normal power at sea level for start engine and taxi, plus one minute of maximum power for take-off, 5 minutes at maximum power at combat ceiling for combat, and a reserve of 20 minutes loiter at sea level at speeds for maximum endurance and 5% of initial fuel load.

FORMULA: POINT-INTERCEPT MISSION III

Take-off and accelerate to best climb speed at maximum power, climb to 45,000 feet with maximum power, combat 5 minutes at 45,000 feet with maximum power, loiter at 20,000 feet at speeds for maximum endurance. Reserve is 10% of initial fuel load. Total mission time does not include 2 minutes at normal power for starting engine and taxi, or reserve.

FORMULA: AREA-INTERCEPT MISSION IV

Take-off and accelerate to best climb speed at maximum power, climb to cruise altitude with military power, cruise at cruise altitude at long range speeds, search at best cruise altitude with normal power for 10 minutes, combat 5 minutes at cruise ceiling with maximum power, cruise back at cruise altitude at long range speeds, and loiter for 10 minutes at 35,000 feet at speeds for maximum endurance. Range-free allowances include 5 minutes at normal power at sea level for start engine and taxi, plus one minute of maximum power for take-off, 5 minutes for combat, loiter 10 minutes at 35,000 feet, and a reserve of 10% of initial fuel load.

FORMULA: RANGE MISSION V

Take-off and accelerate to best climb speed with maximum power, climb to cruise altitude with military power, cruise at cruise altitude at long range speeds to remote base. Range-free allowances include 5 minutes at normal power at sea level for start engine and taxi plus one minute of maximum power for take-off, and a reserve of 20 minutes loiter at sea level at speeds for maximum endurance and 5% of initial fuel load.

GENERAL DATA:

- (a) Tanks dropped when empty unless otherwise specified.
- (b) For detailed planning refer to Tech Order IF-86L-1 and other applicable technical orders.
- (c) Engine ratings shown on page 3 are manufacturer's guaranteed values. Power values used in performance calculations are as follows:

(1) J47-GE-33		
S. L. Static	LB	TEMP
Max:	*6830	685°C
Mil:	5080	685°C
Nor:	4680	635°C

*With afterburner operating

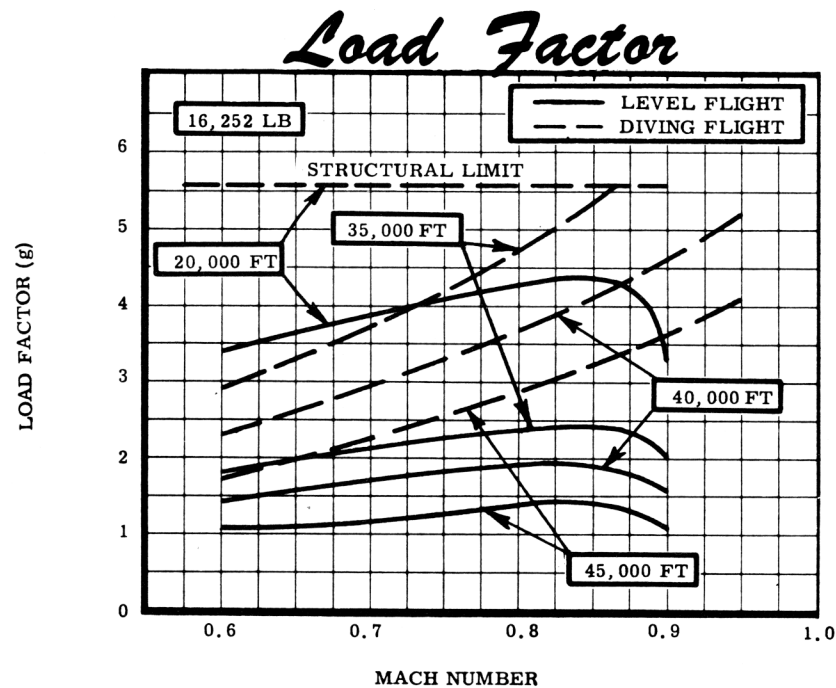
7950 RPM remains constant for normal power and higher.

PERFORMANCE BASIS:

North American Report Nr NA-54-781, Appendix V, "Performance Calculations based on Flight Test Results of Service Model F-86D Airplane", dated 5 July 1956.

REVISION BASIS:

To revise Electronics and Fuel blocks.

SUPPLEMENTAL

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